



Coupled Field Simulation

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Austria

❑ Overview about CFS++

- History
- Capabilities
- General Approach

❑ Preprocessing

- Ansys (classic), Gmesh, ..
- Physical definitions
- Material definition

❑ Performing simulation

- Simulation parameters

❑ Postprocessing

- Result files
- Paraview

❑ Starting

- M.Kaltenbacher and PhD students at Department of Sensor Technology, Friedrich-Alexander University Erlangen-Nuremberg, year 2000

❑ Current involved groups

- Kaltenbacher's group

- Institute of Mechanics and Mechatronics, Vienna University of Technology



- R. Lerch's group

- Department of Sensor Technology, Friedrich-Alexander University Erlangen-Nuremberg



- M. Stingl's group

- Chair of Applied Mathematics 2, Friedrich-Alexander University Erlangen-Nuremberg



❑ Co-operation partners

- S. Reitzinger (support in the design phase)
- M. Hofer (bilinear and linear forms)
- M. Mohr (algebraic system and XML-structure)
- T. Lahmer (inverse schemes)
- B. Schmidt (contact handling)

❑ Former Ph.D. students

- E. Zhelezina (first implementations)
- M. Escobar (aeroacoustics)
- G. Link (fluid flow and coupling to mechanics)

❑ Current Ph.D. students

- A. Hauck (revisions of class structure, p-FEM, gFEM)
- J. Grabinger (non-matching grids, aeroacoustics)
- S. Zörner (fluid-structure-acoustic)
- A. Volk (magnetostriction)

❑ Post Docs and others

- A. Hüppe (aeroacoustics, spectral-FEM, mixed FEM, revisions of class structure)
- M. Kaltenbacher (nonlinear PDEs, moving mesh, material modeling, hysteresis)
- S. Triebenbacher (non-matching grids, aeroacoustics, CFSDEPS, CMake)
- F. Wein (optimization)

❑ Capabilities

➤ Single Field:

- Electromagnetics
- Mechanics
- Acoustics
- Flow

➤ Multi-Field

- Piezoelectricity
- Electrostatics-Mechanics-Acoustics
- Magnetic-Mechanics-Acoustics
- Flow-Mechanics-Acoustics



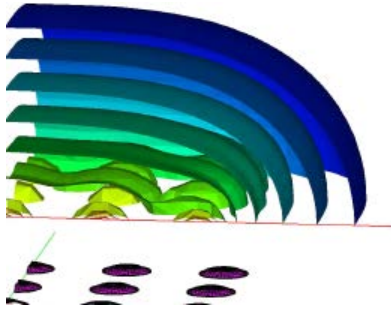
➤ FE Topics

- **Finite Element of higher order**
(pFEM, spectral FEM)
- **None-conforming grids**
- Moving meshes
- **Material models with hysteresis**
- Micro-Macro Modeling (Homogenization)
- Fast solvers (AMG, Pardiso)
- **Optimization (SIMP, topology, shape)**
- **Inverse schemes (material parameter determination)**

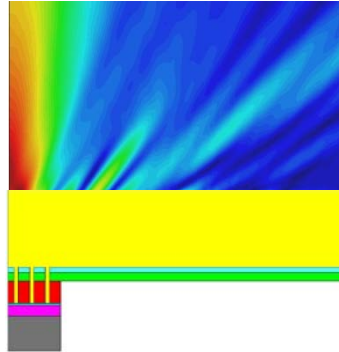


CFS++ Capabilities

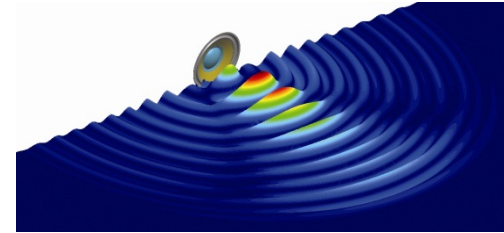
CMUTs



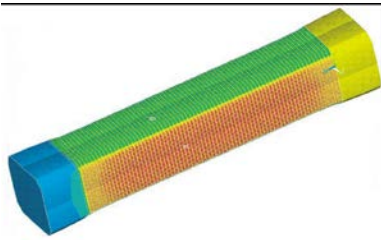
Piezoelectric US-array



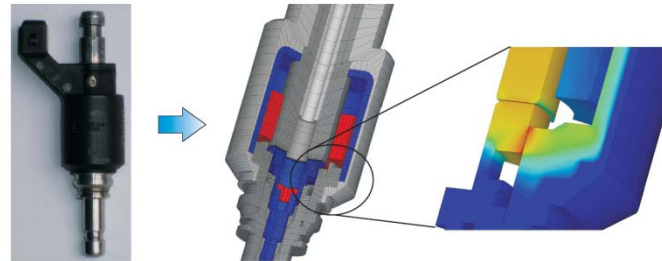
Electrodynamic loudspeaker



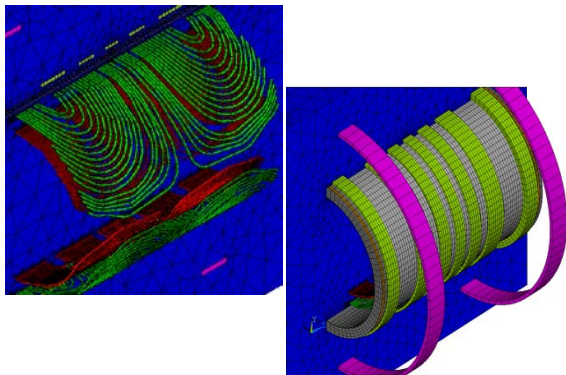
Piezoelectric stack



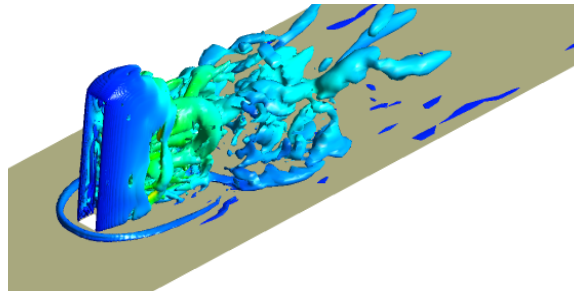
Electromagnetic valve



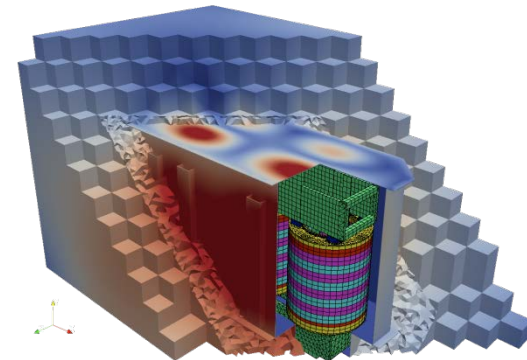
MRI-Scanner



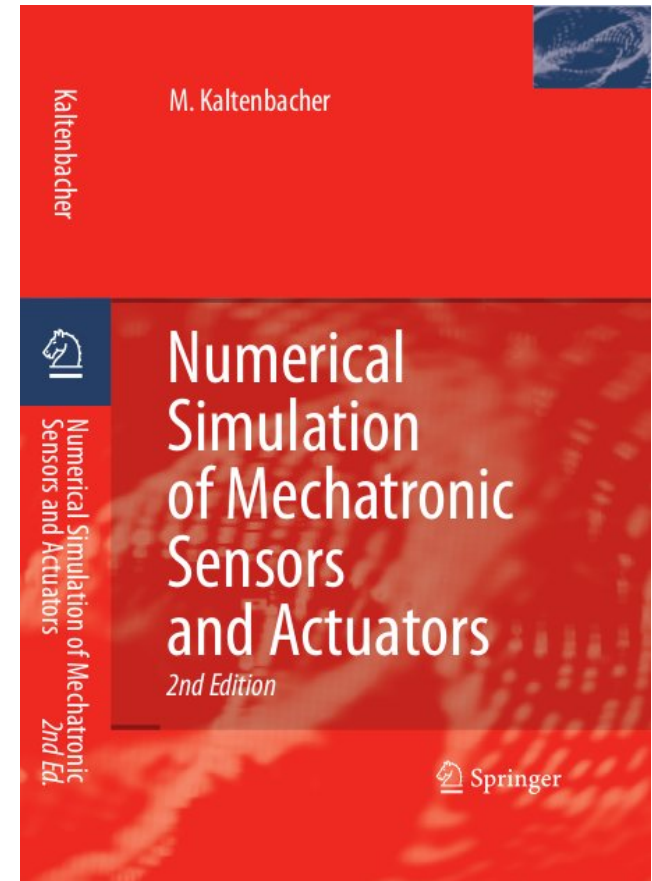
Flow-induced sound



Transformer noise



- ❑ M. Kaltenbacher: *Numerical Simulation of Mechatronic Sensors and Actuators*, 2nd Edition, Springer 2007, 2nd edition
- ❑ M. Kaltenbacher: Advanced Simulation Tool for the Design of Sensors and Actuator. Proceeding of Eurosensors XXIV 2010
- ❑ Just search for scientific articles
 - Andreas Hauck
 - Andreas Hüppe
 - Jens Grabinger
 - Tom Lahmer
 - Gerhard Link
 - Manfred Kaltenbacher
 - Simon Triebenbacher
 - Adiran Volk
 - Fabian Wein
 - Stefan Zörner
 -

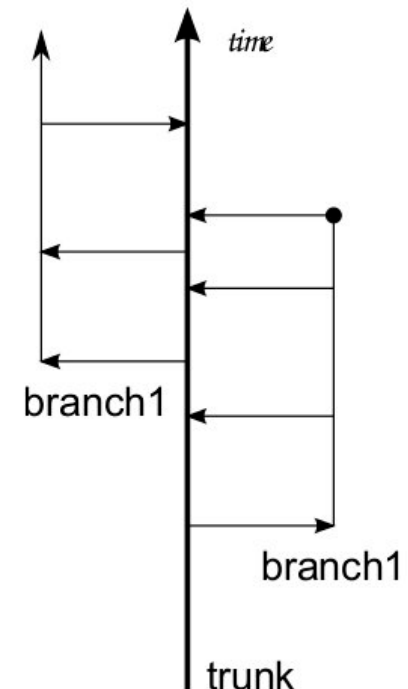


❑ Overview

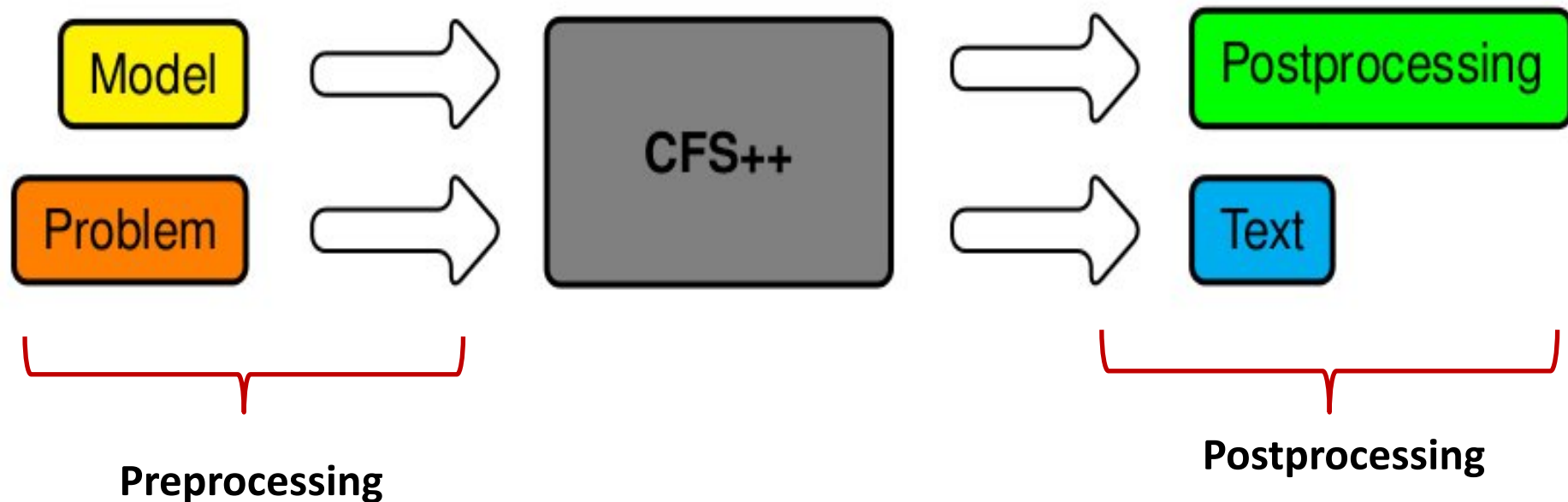
- No license file exists!
- CFS++ is considered open source as it has been developed, and is still developed by projects funded by public research agencies!

❑ Gentleman's agreement

- CFS++ shall not be publicly available
- Manfred Kaltenbacher is informed prior to distribution
- Currently: svn-server in Erlangen
- Development approach
 - Everybody works on own branches
 - Test cases prevent side effects
 - Not everything ends on the trunk



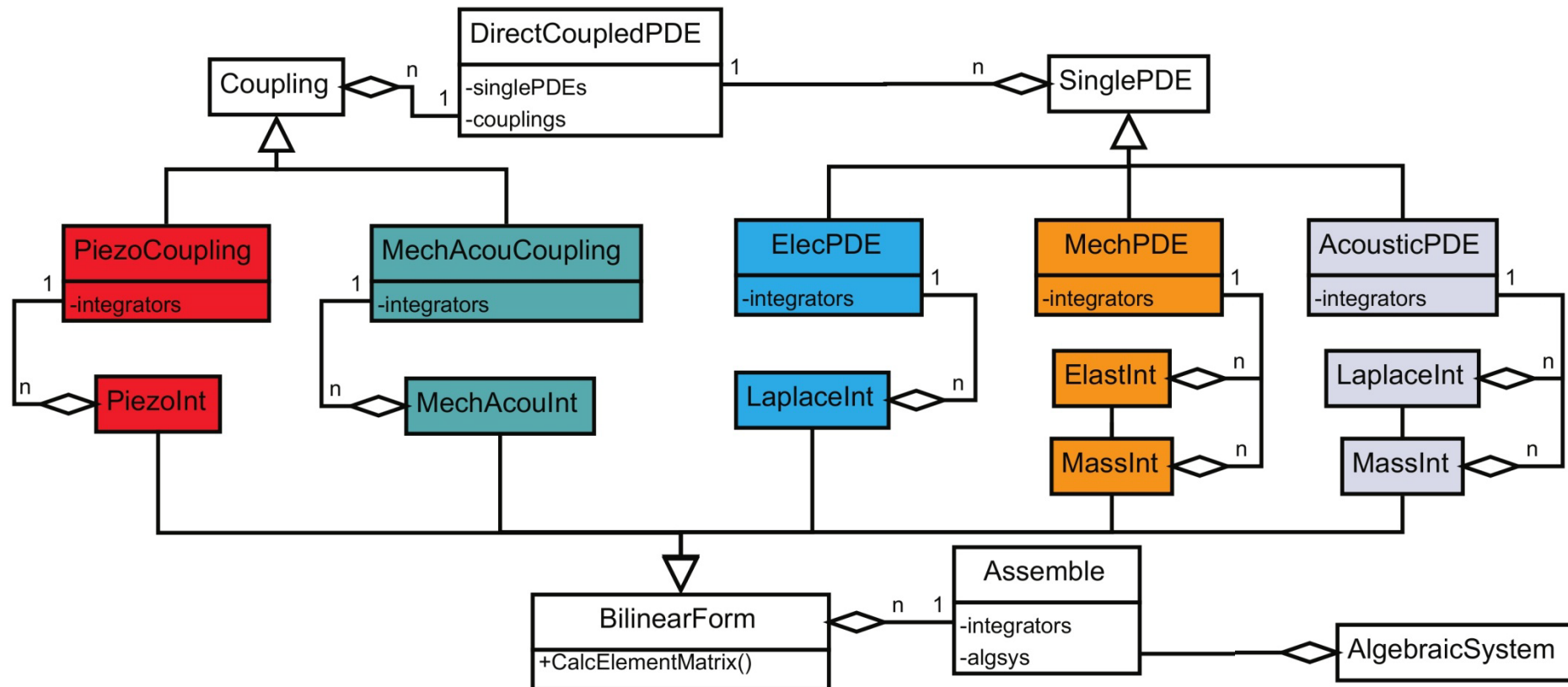
- ❑ **Model:** mesh data
- ❑ **Problem:** XML file defining problem and physical properties
- ❑ **CFS++:** command line based FEM solver/ optimizer
- ❑ **Postprocessing:** visualization and calculations
- ❑ **Text:** detailed information in XML format



CFS++: General Approach

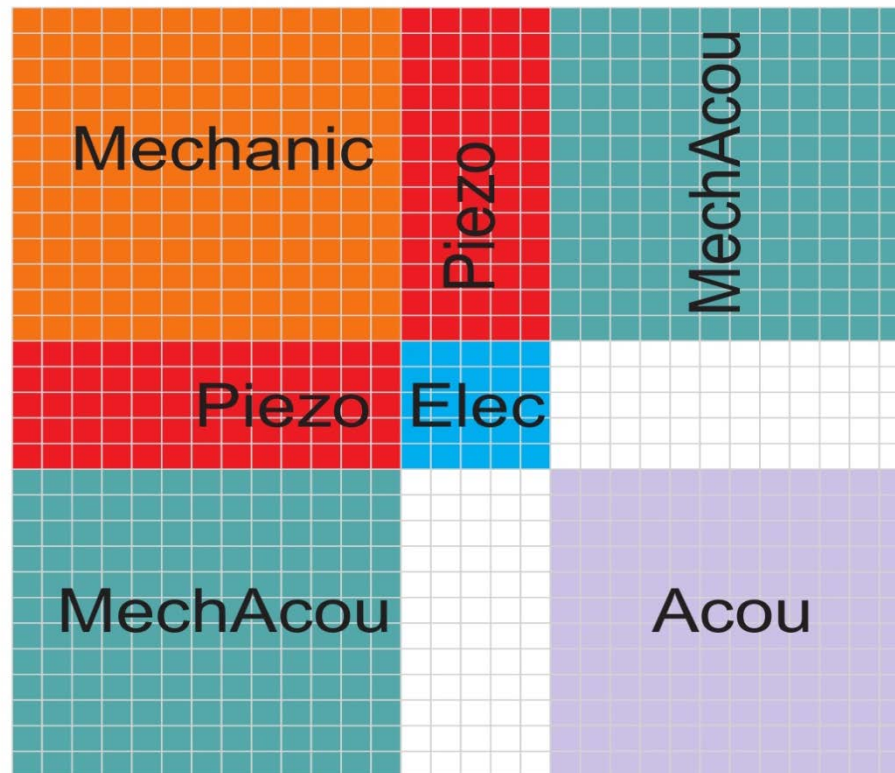
❑ CFS++ (Coupled Field Simulation):

- Models: Based on partial differential equations (PDEs)
- Solution of PDEs: Finite-Element-Method
- Example: Structure for piezoelectric-acoustic coupled PDEs



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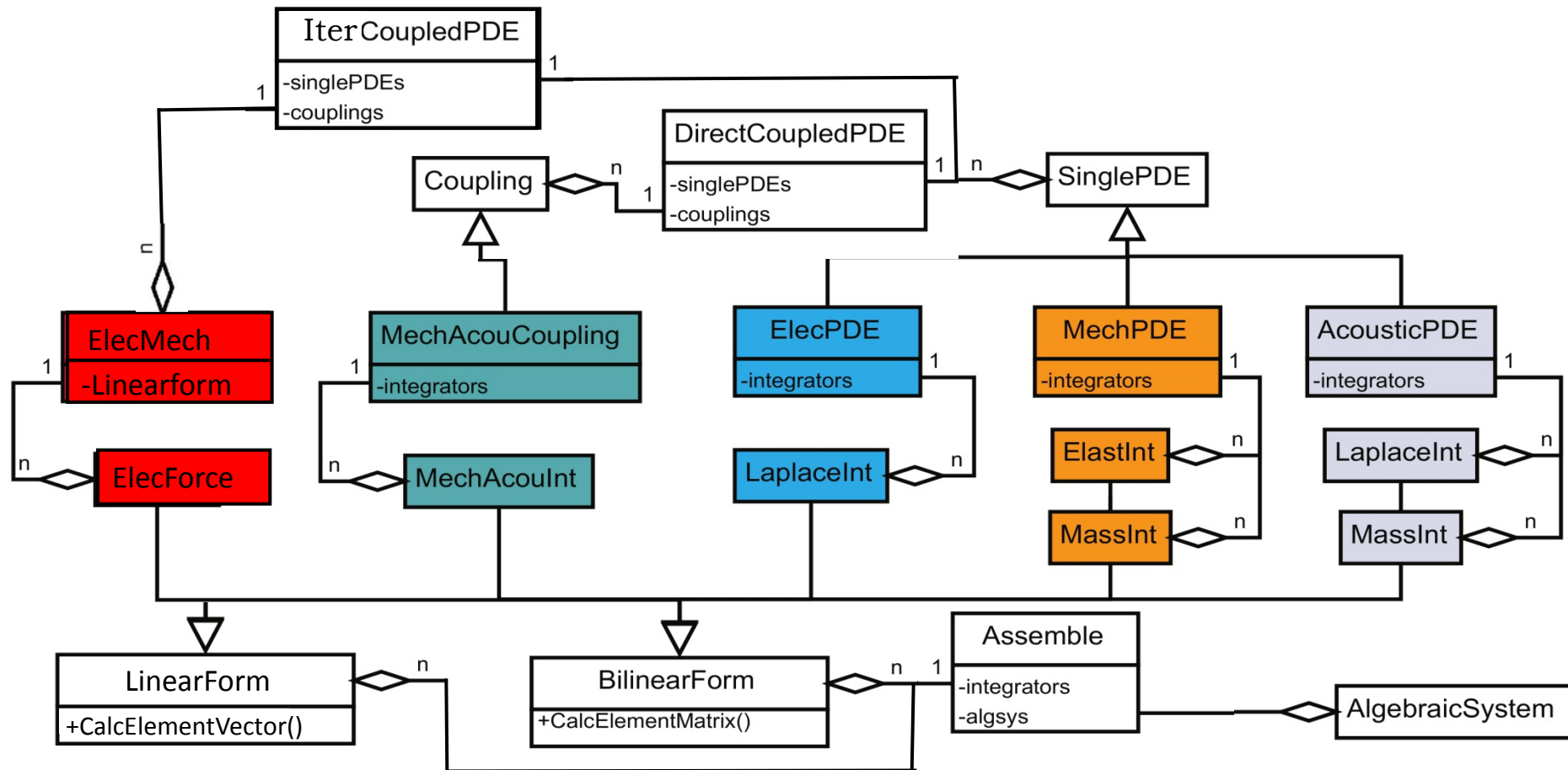
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CFS++: General Approach

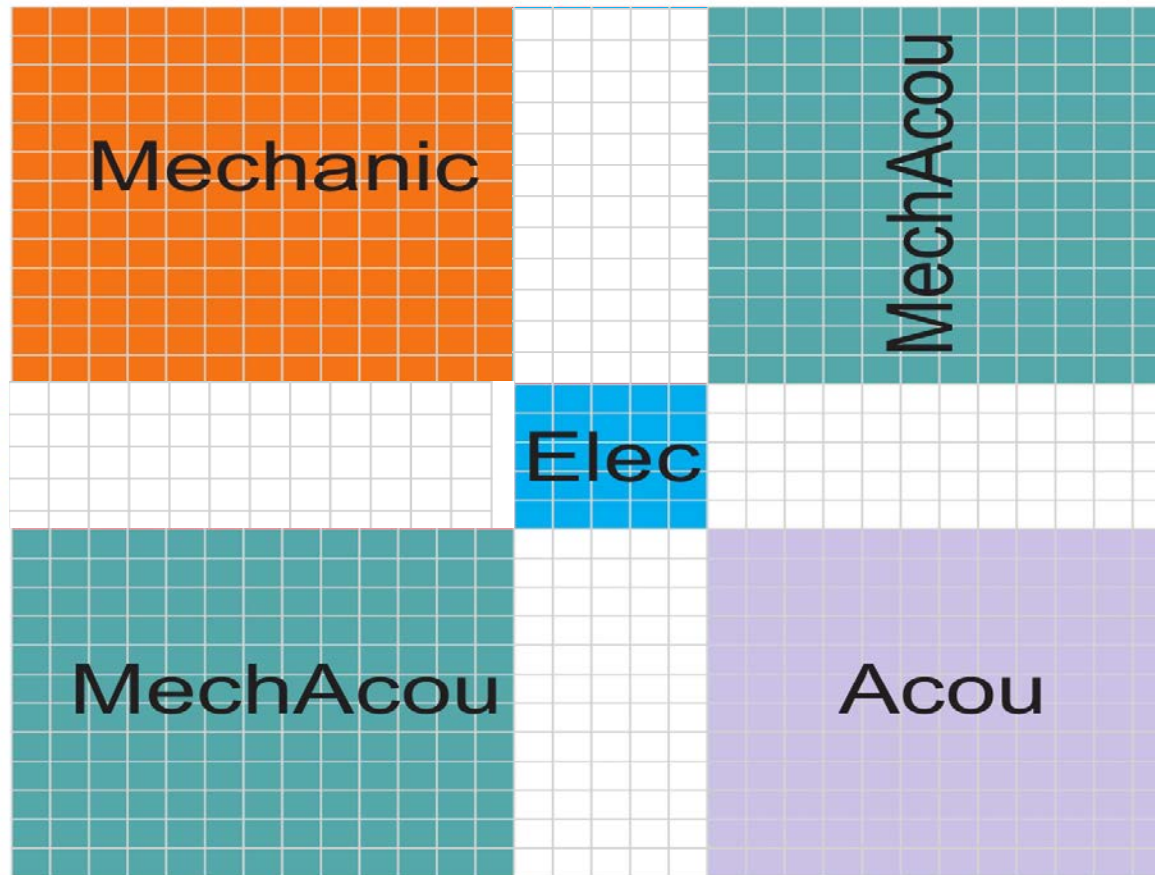
❑ CFS++ (Coupled Field Simulation):

➤ Coupling: Electrostatics-Mechanics-Acoustics (MEMS)



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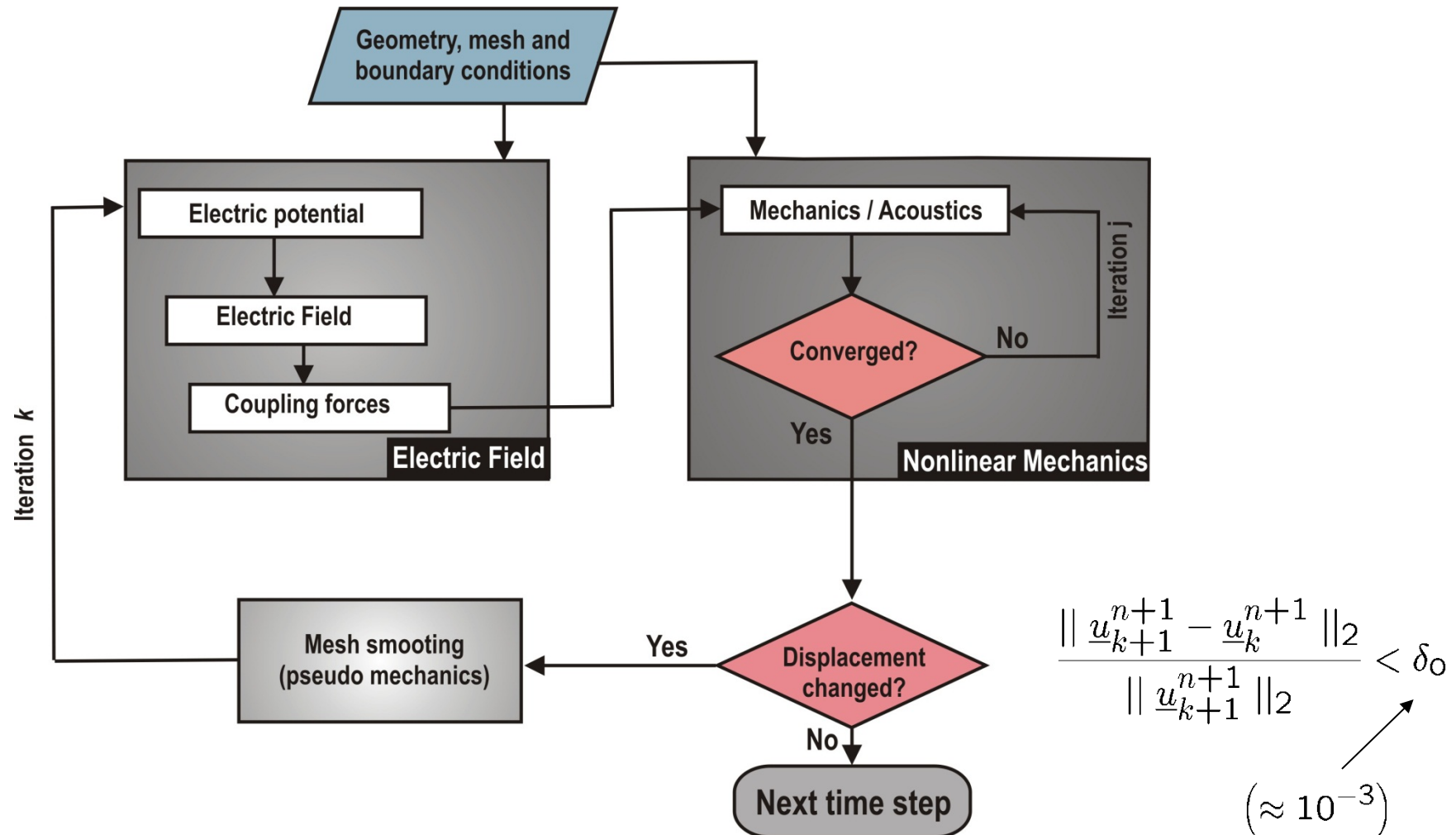
- Coupling: Electrostatics-Mechanics-Acoustics (MEMS)



CFS++: General Approach

❑ CFS++ (Coupled Field Simulation):

➤ Coupling: Electrostatics-Mechanics-Acoustics (MEMS)



❑ Pre-processing

- Supported pre-processors: ANSYS, GiD, Gmesh, OpenFoam, ANSYS-CFX, ...



Coordinates and mesh topology of all 3D (volume), 2D (surface) and 1D (interface) elements

❑ Physical problem description: XML-format

```
1  <couplingList>
2    <direct>
3      <piezoDirect>
4        <regionList>
5          <region name="solid"/>
6        </regionList>
7      </piezoDirect>
8      <acouMechDirect>
9        <surfRegionList>
10         <surfRegion name="interface"/>
11        </surfRegionList>
12      </acouMechDirect>
13    </direct>
14  </couplingList>
```

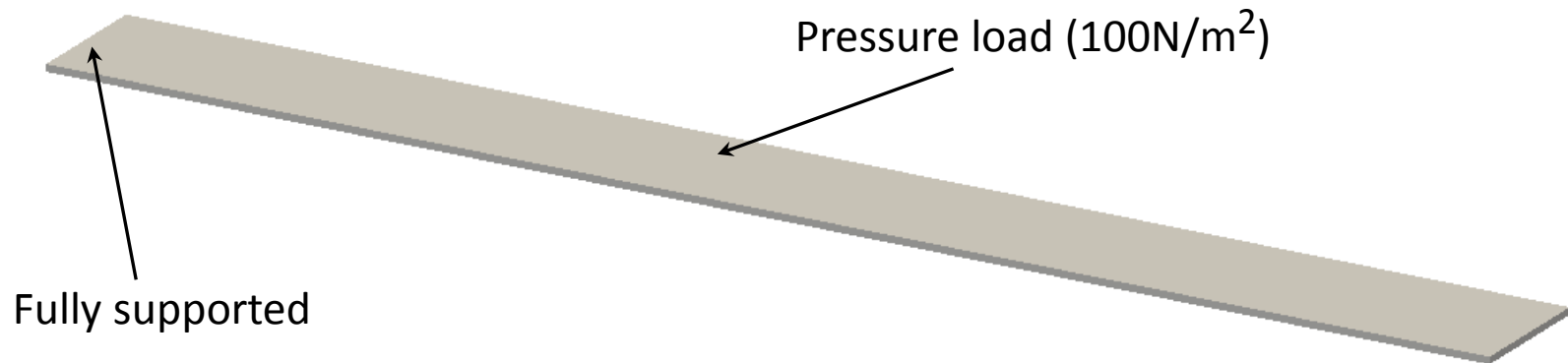
❑ Post-processing

- Supported post-processors: ANSYS, GiD, Gmesh, GMV, VTK, ParaView, ..

Example 1: Mechanics

❑ Problem setup

- Cantilever with pressure load



❑ Goal

- Perform
 - Static analysis
 - Transient analysis
 - Harmonic analysis
 - Eigenfrequency analysis

Example 1: Mechanics

❑ Preprocessor: Ansys classic, use prep7 commands

➤ Initialization part

```
fini  
/uis,msgpop,3 ! disable warning popups  
/clear  
/filename,cantilever,1 ! important: define name of the model  
/prep7  
/uis,msgpop,2 ! enable warning popups  
init ! initialize macros for mesh-interface to CFS++
```

➤ Define constants and generate geometry

```
!define constants  
eps=1e-7  
L=400e-6  
b=2e-6  
bc=45e-6  
!generate the cantilever  
block,0,L,0,bc,0,b
```

Example 1: Mechanics

❑ Preprocessor: Ansys classic, use prep7 commands

➤ Perform volume meshing

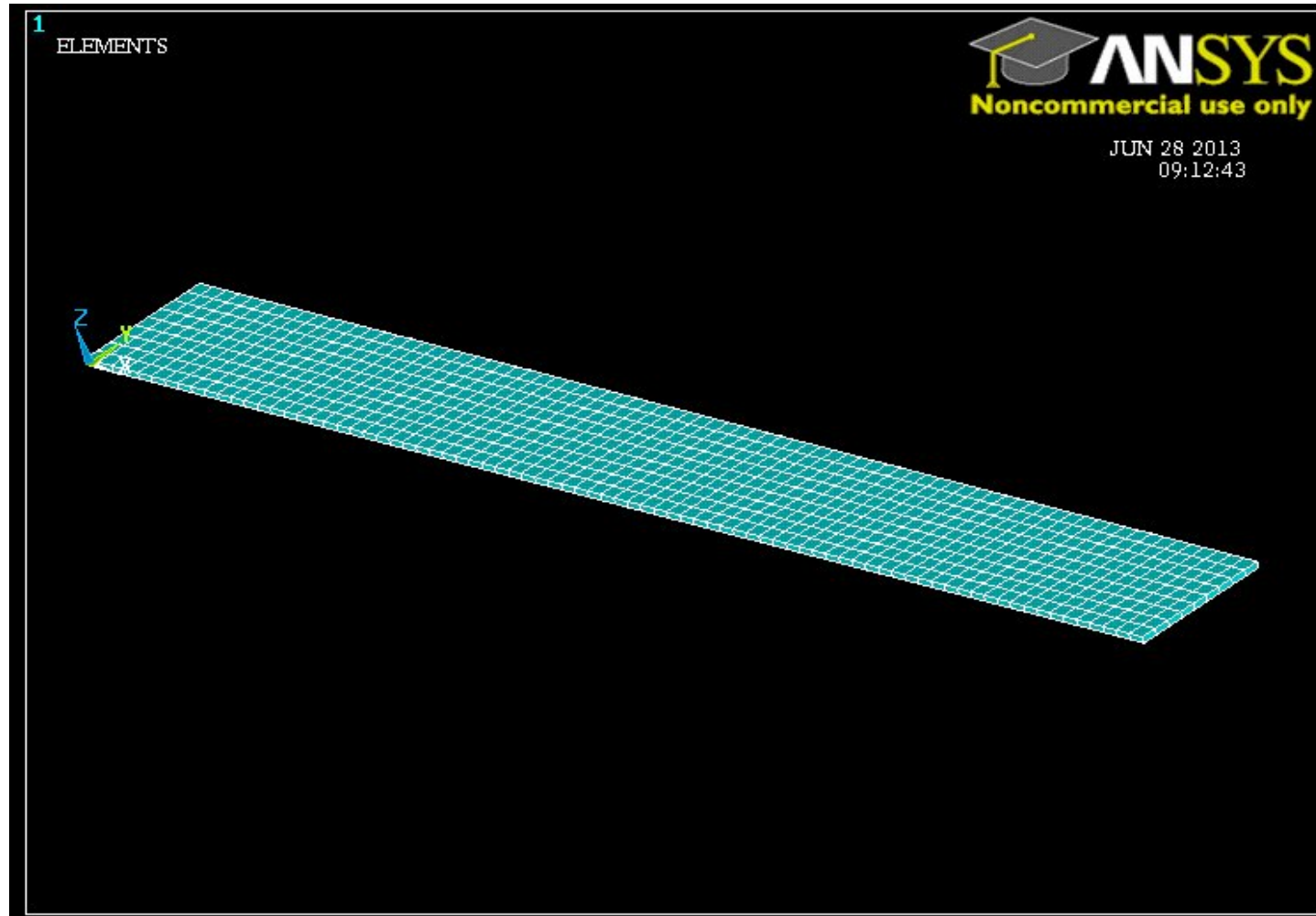
```
setelems,'brick'  
esize,5e-6  
vsel,s,,,1  
vmesh,all
```

➤ Perform surface mesh for pressure load and mechanical fixing

```
setelems,'quadr'  
allsel,  
asel,s,,,2  
amesh,all  
allsel  
asel,s,loc,x,-eps,eps  
asel,r,loc,y,-eps,bc+eps  
asel,r,loc,z,-eps,eps+b  
amesh,all
```

Example 1: Mechanics

- ❑ Preprocessor: Ansys classic, use prep7 commands
 - Display of mesh



Example 1: Mechanics

❑ Preprocessor: Ansys classic, use prep7 commands

➤ Write mesh file for CFS++: Write volume elements

```
allsel,all  
vsel,s,,,1  
eslv  
welems,'cantilever'
```

➤ Write mesh file for CFS++: Write surface elements

```
!write surface elements for pressure load  
allsel  
asel,s,,,2  
esla  
welems,'excite'  
!write surface elements for mechanical fixing  
allsel  
asel,s,loc,x,-eps,eps  
asel,r,loc,y,-eps,bc+eps  
asel,r,loc,z,-eps,eps+b+b  
esla  
welems,'fix-all'
```

Example 1: Mechanics

❑ Write mesh file for CFS++

- Write mesh file for CFS++: Write coordinates of nodes

```
! write all nodes  
allsel  
wnodes
```

- Write mesh file for CFS++: Write the whole hdf5-file

```
! write hdf5-mesh file  
mkhdf5
```

- Result

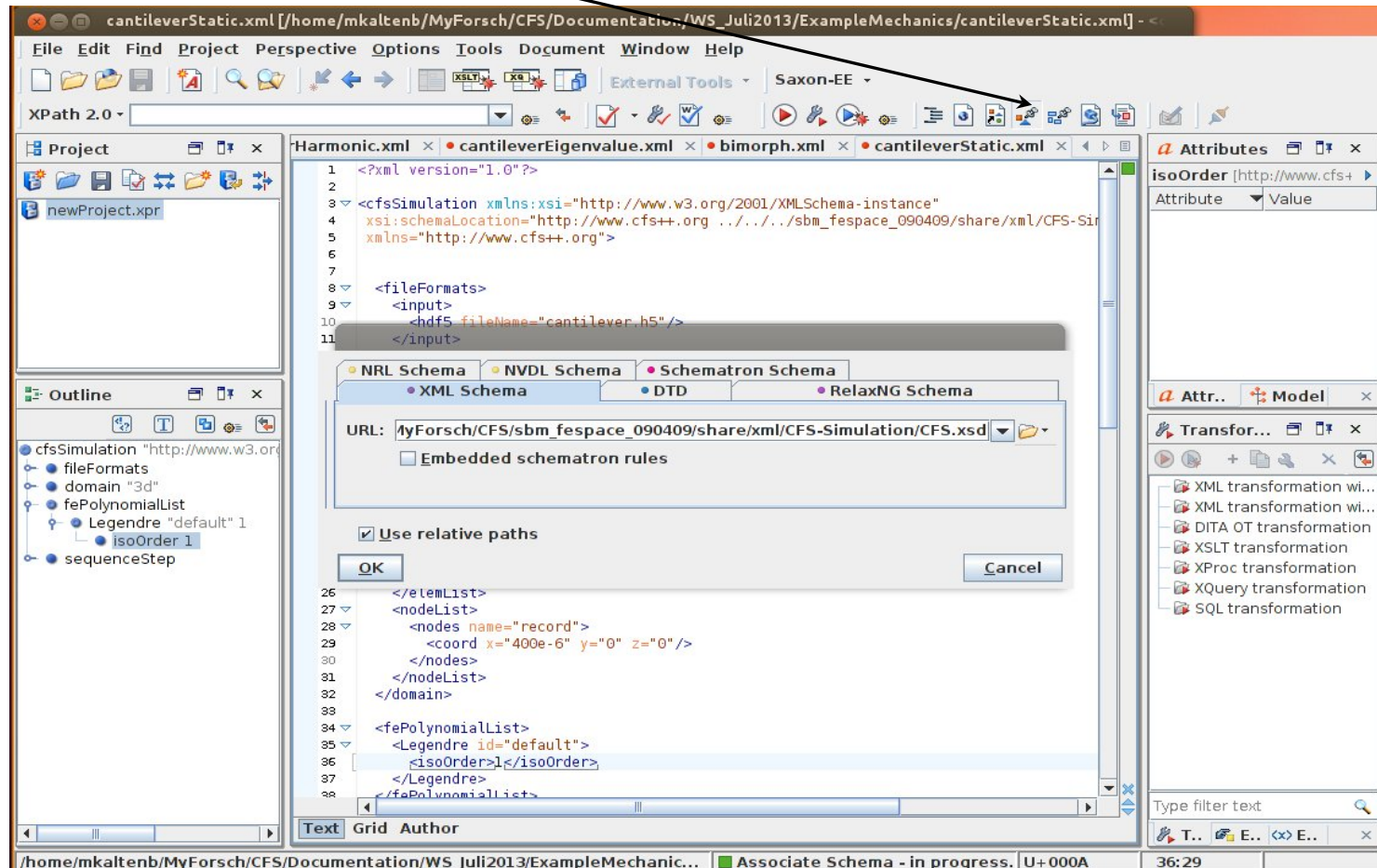
- Within this directory you should now find the file **cantilver.h5!**

❑ Problems:

- If there are inconsistencies, then you will get the error „MKHDF5: Please check the contents of cantilever_cpreader.out”.

Example 1: Mechanics

- ❑ Setup the xml-File for describing the physics and simulation parameter
 - Use oxygen (see <http://www.oxygenxml.com/>)
 - Load the correct schema



Example 1: Mechanics

❑ Setup the xml-File for describing the physics and simulation parameter

- Hint: search for a similar problem in the CFS++ - TESTSUIT
- Header

```
<?xml version="1.0"?>  
<cfsSimulation xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"  
xsi:schemaLocation=http://www.cfs++.org/..../sbm\_fespace\_090409/share/xml/CFS-Simulation/CFS.xsd xmlns="http://www.cfs++.org">
```

- File section

```
<fileFormats>  
  <input>  
    <hdf5 fileName="cantilever.h5"/>  
  </input>  
  <output>  
    <text/>  
    <hdf5/>  
  </output>  
  <materialData file="mat.xml"/>  
</fileFormats>
```

Example 1: Mechanics

- ❑ Setup the xml-File for describing the physics and simulation parameter
 - Define computational domain with volume-regions, surface-regions, ..

```
<domain geometryType="3d">  
  <regionList>  
    <region name="cantilever" material="Steel"/>  
  </regionList>  
  <surfRegionList>  
    <surfRegion name="excite"/>  
    <surfRegion name="fix-all"/>  
  </surfRegionList>  
  <nodeList>  
    <nodes name="record">  
      <coord x="400e-6" y="0" z="0"/>  
    </nodes>  
  </nodeList>  
</domain>
```

Define a new node for data recording

Example 1: Mechanics

❑ Setup the xml-File for describing the physics and simulation parameter

- Start the sequence and define the analysis type

```
<sequenceStep>  
  <analysis>  
    <static/>  
  </analysis>
```

- Define the partial differential equation (PDE) and the region it is defined

```
<pdeList>  
  <mechanic subType="3d">  
    <regionList>  
      <region name="cantilever"/>  
    </regionList>
```

Example 1: Mechanics

❑ Setup the xml-File for describing the physics and simulation parameter

➤ Define boundary consitions

```
<bcAndLoads>
  <pressure name="excite" value="100" />
  <fix name="fix-all">
    <comp dof="x"/>
    <comp dof="y"/>
    <comp dof="z"/>
  </fix>
</bcAndLoads>
```

➤ Define store results

```
<storeResults>
  <nodeResult type="mechDisplacement">
    <allRegions/>
    <nodeList>
      <nodes name="record"/>
    </nodeList>
  </nodeResult>
</storeResults>
```

Example 1: Mechanics

❑ Setup the xml-File for describing the physics and simulation parameter

➤ Close PDE part

```
</mechanic>  
</pdeList>
```

➤ Define solver (not necessary)

```
<linearSystems>  
  <system>  
    <solverList>  
      <pardiso>  
    </pardiso>  
    </solverList>  
  </system>  
</linearSystems>
```

➤ Close the sequence and the whole xml-definition

```
</sequenceStep>  
</cfsSimulation>
```

Example 1: Mechanics

- ❑ Setup the xml-File for the material parameters

```
<?xml version="1.0"?>

<cfsMaterialDataBase xmlns="http://www.cfs++.org">
  <material name="Steel">
    <mechanical>
      <density> 7.8E03 </density>
      <elasticity>
        <isotropic>
          <real>
            <elasticityModulus> 210E09 </elasticityModulus>
            <poissonNumber> 0.3 </poissonNumber>
          </real>
        </isotropic>
      </elasticity>
    </mechanical>
  </material>
</cfsMaterialDataBase>
```

Example 1: Mechanics

❑ Perform the simulation with CFS++

- Just „cfs“ you get the following help (provided that the executable is in the path)

```
=====
CFS++ - Coupled Field Simulation

v. 13.06 - 'Nonmatching Noolbenger' (rev 12495 modified)
compiled Jun 24 2013 as Release
=====

Usage: cfs [param] <name of simulation run>

where [param] is one or more of the following parameters:
-h [ --help ]           display this usage information
-v [ --version ]        information about the CFS++ executable
-H [ --history ]        history of revisions
-m [ --meshFile ] arg   name of mesh file for the simulation
-p [ --paramFile ] arg  name of XML parameter file for the simulation
-s [ --schemaRoot ] arg path to XML schema definitions (env CFS_SCHEMA_ROOT)
-x [ --ersatz ] arg     name of ersatz material density file
-r [ --restart ]         read restart file of previous simulation run
-f [ --forceSegFault ]  force a segmentation fault at exceptions
-g [ --printGrid ]      read grid from input and write it to output file
-G [ --exportGrid ]     export the grid to the info.xml file, best with -g
-w [ --writeSkeleton ]  write skeleton of XML file for subsequent simulation
-l [ --logConfFile ] arg name of configuration file for logging streams
-d [ --detailed ]       detailed output to info.xml
-q [ --quiet ]          more compressed console output (env CFS_QUIET)
--noColor              turn off colored output
```

Example 1: Mechanics

❑ Perform the simulation with CFS++

➤ Type in „cfs cantileverStatic“ and you obtain

```
=====
CFS++ - Coupled Field Simulation

v. 13.06 - 'Nonmatching Noolbenger' (rev 12495 modified)
compiled Jun 24 2013 as Release
=====

Simulation run started at 2013-Jun-28 10:34:08 on nb1
++ Reading parameter file '/home/mkaltenb/MyForsch/CFS/Documentation/WS_Juli2013/ExampleMechanics/cantileverStatic.xml'
++ Creating HDF5 writer 'hdf5'
++ Reading mesh ... "cantilever.h5" -> regular
++ Creating PDE 'mechanic'
++ Starting to solve problem

++ Finished solving the problem at 2013-Jun-28 10:34:08

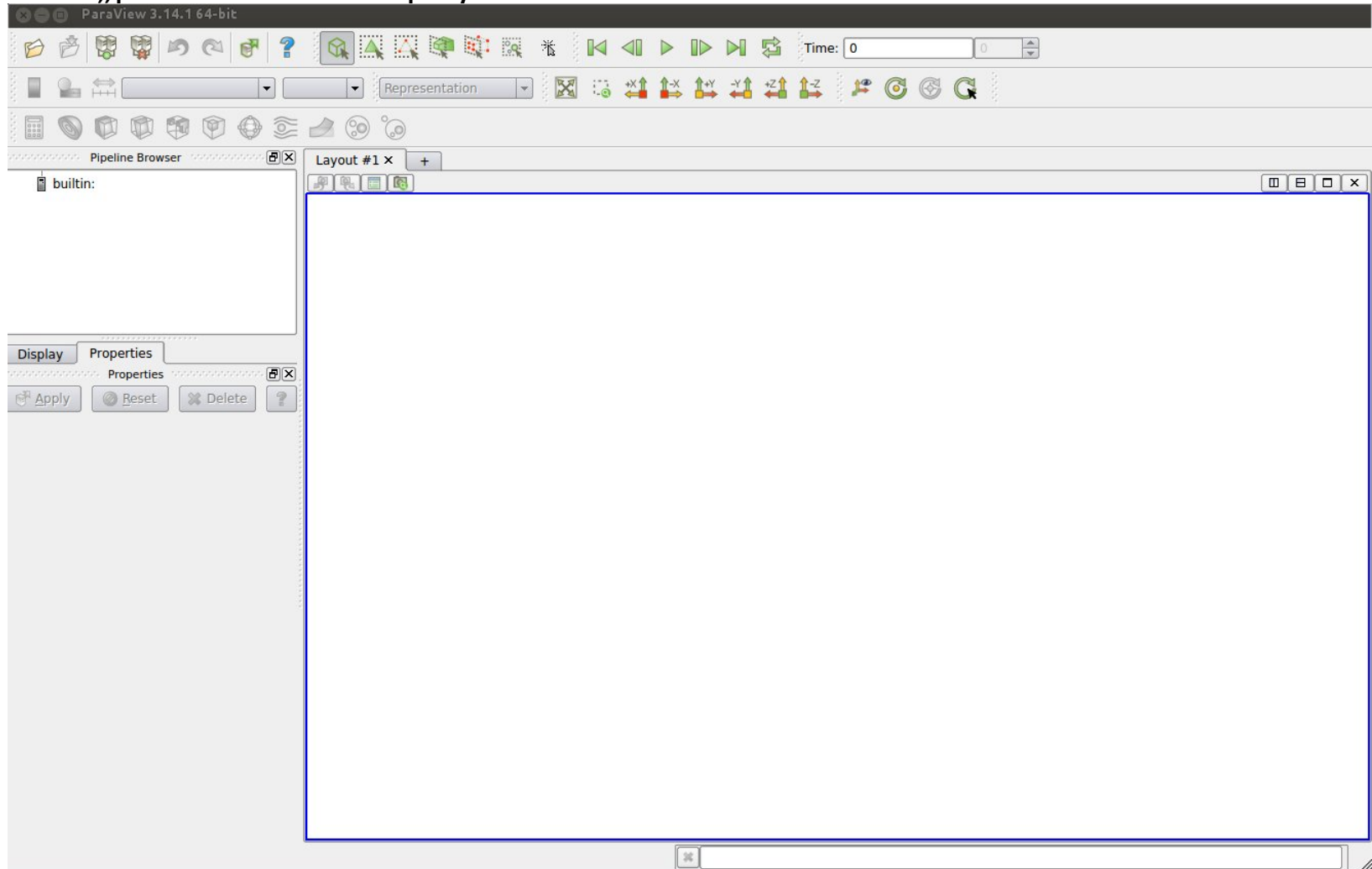
>> Total time: wall clock: '0s' CPU time: '0s'
```

➤ Generated output files

- In results_hdf5: cantileverStatic.h5: contains the simulation result
- In history: cantileverStatic-ms1-mechDisplacement-node-91-record.hist
- In current directory: cantileverStatic.info.xml

Example 1: Mechanics

- ❑ Perform postprocessing with paraview (<http://www.paraview.org/>)
 - Start „paraview“ and display the result



Example 1: Mechanics

❑ Perform a transient analysis

➤ Change analysis part

```
<sequenceStep>  
  <analysis>  
    <transient>  
      <numSteps> 40 </numSteps>  
      <deltaT> 0.05 </deltaT>  
    </transient>  
  </analysis>
```

➤ Change in „bcsAndLoads“ for pressure excitation

```
<bcsAndLoads>  
  <pressure name="excite" value="100*sin(2*pi*1*t" />
```

Frequency of 1Hz

Example 1: Mechanics

❑ Perform a harmonic analysis

➤ Change analysis type

```
<sequenceStep>  
  <analysis>  
    <harmonic>  
      <numFreq> 10 </numFreq>  
      <startFreq> 10 </startFreq>  
      <stopFreq> 1e4 </stopFreq>  
      <sampling>linear</sampling>  
    </harmonic>  
  </analysis>
```

➤ Change pressure load towards amplitude and phase

```
<bcsAndLoads>  
  <pressure name="excite" value="100" phase="0.0" />
```

Example 1: Mechanics

❑ Perform a eigenfrequency analysis

➤ Change analysis type

```
<sequenceStep>
  <analysis>
    <eigenFrequency>
      <isQuadratic> no </isQuadratic>
      <numModes> 5 </numModes>
      <freqShift> 0 </freqShift>
      <writeModes> yes </writeModes>
    </eigenFrequency>
  </analysis>
```

➤ No load is needed (not necessarily be changed)

```
<bcAndLoads>
  <fix name="fix-all">
    <comp dof="x"/>
    <comp dof="y"/>
    <comp dof="z"/>
  </fix>
</bcAndLoads>
```

Example 1: Mechanics

❑ Perform the simulation with CFS++

➤ Type in „cfs cantileverEigenfrequency“ and you obtain

```
=====
CFS++ - Coupled Field Simulation

v. 13.06 - 'Nonmatching Noolbenger' (rev 12495 modified)
compiled Jun 24 2013 as Release
=====

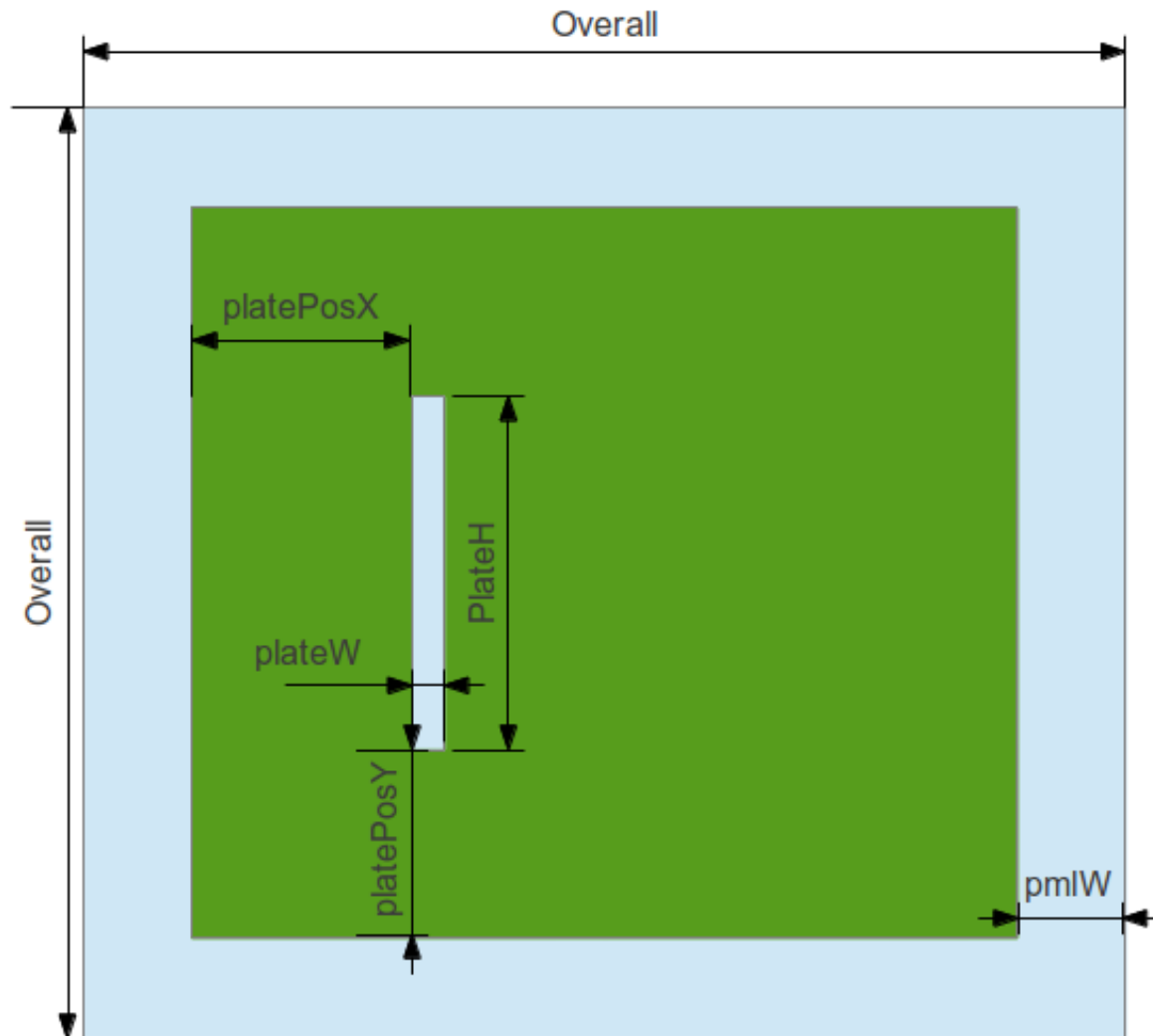
Simulation run started at 2013-Jun-28 11:03:55 on nb1
++ Reading parameter file '/home/mkaltenb/MyForsch/CFS/Documentation/
++ Creating HDF5 writer 'hdf5'
++ Reading mesh ... "cantilever.h5" -> regular
++ Creating PDE 'mechanic'
++ Starting to solve problem

      Frequency [Hz] |      Errorbound
-----|-----
      20075.6 |      6.79268e-14
      125806 |      3.16438e-15
      187699 |      2.0478e-15
      234863 |      1.82875e-15
      352345 |      6.26307e-13

++ Finished solving the problem at 2013-Jun-28 11:03:55
>> Total time: wall clock: '0s' CPU time: '0s'
```

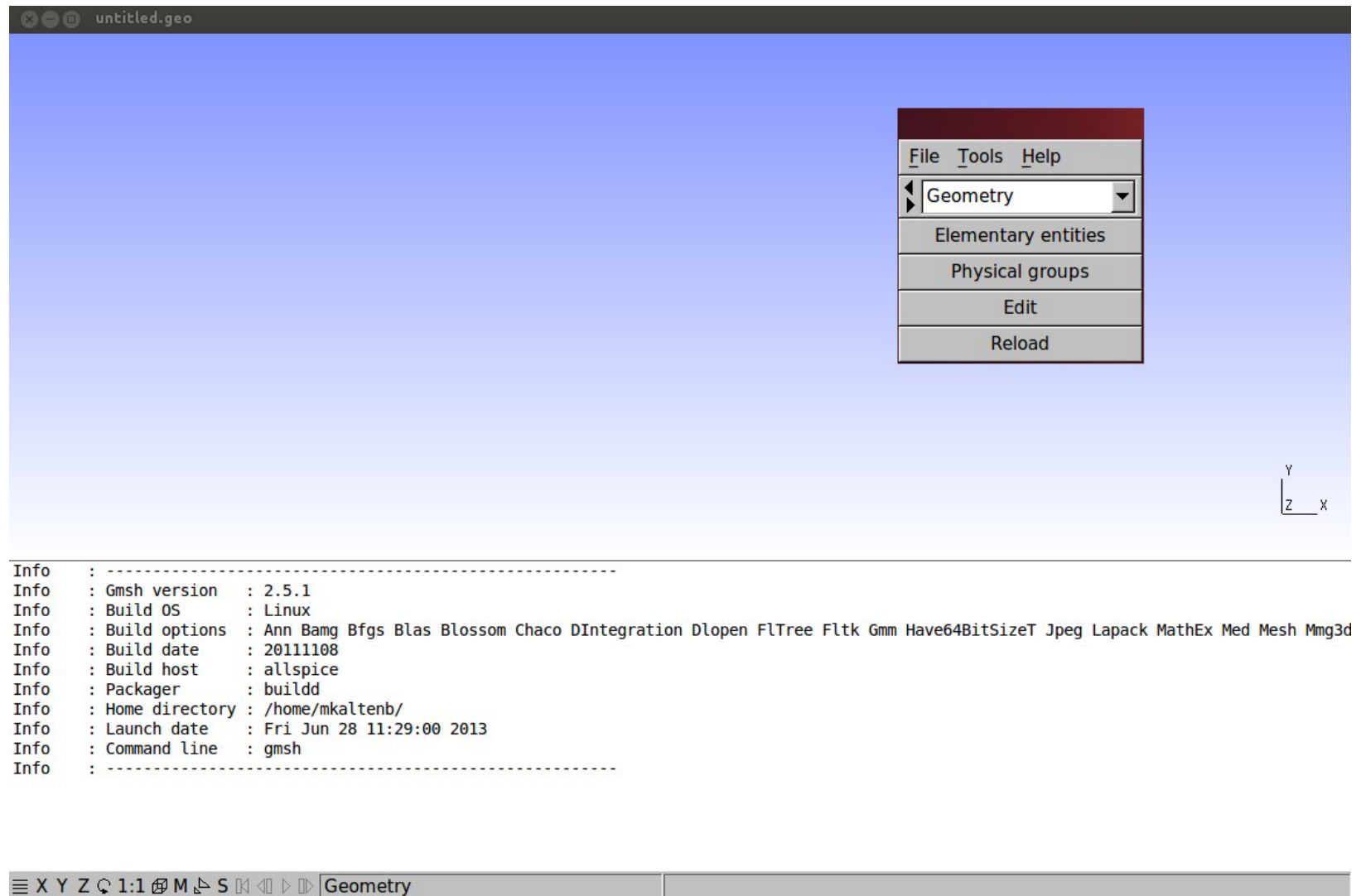
Example 2: Acoustics

- ❑ Problem setup: wave propagation with sound barrier (point source excitation)



Example 2: Acoustics

❑ Preprocessor: gmsh (<http://geuz.org/>)



Example 2: Acoustics

❑ Define constants

```
overall = 10;  
pmlW = 1;  
plateW = 0.1;  
plateH = 3.5;  
platePosX = 2;  
platePosY = 2.25;  
elemSize = 0.1;
```

❑ Bottom up modeling approach for structured mesh:

➤ points->lines->surfaces->mesh

```
//define basepoint for plate  
basePoint = -overall/2+pmlW+platePosY;  
Point(1) = {-overall/2, -overall/2 , 0, 1.0};  
Point(2) = {-overall/2, -overall/2+pmlW , 0, 1.0};  
Point(3) = {-overall/2, basePoint , 0, 1.0};  
Point(4) = {-overall/2, basePoint+plateH , 0, 1.0};  
Point(5) = {-overall/2, overall/2-pmlW , 0, 1.0};  
Point(6) = {-overall/2, overall/2 , 0, 1.0};
```

Example 2: Acoustics

❑ Define lines

```
Line(1) = {1, 2};
```

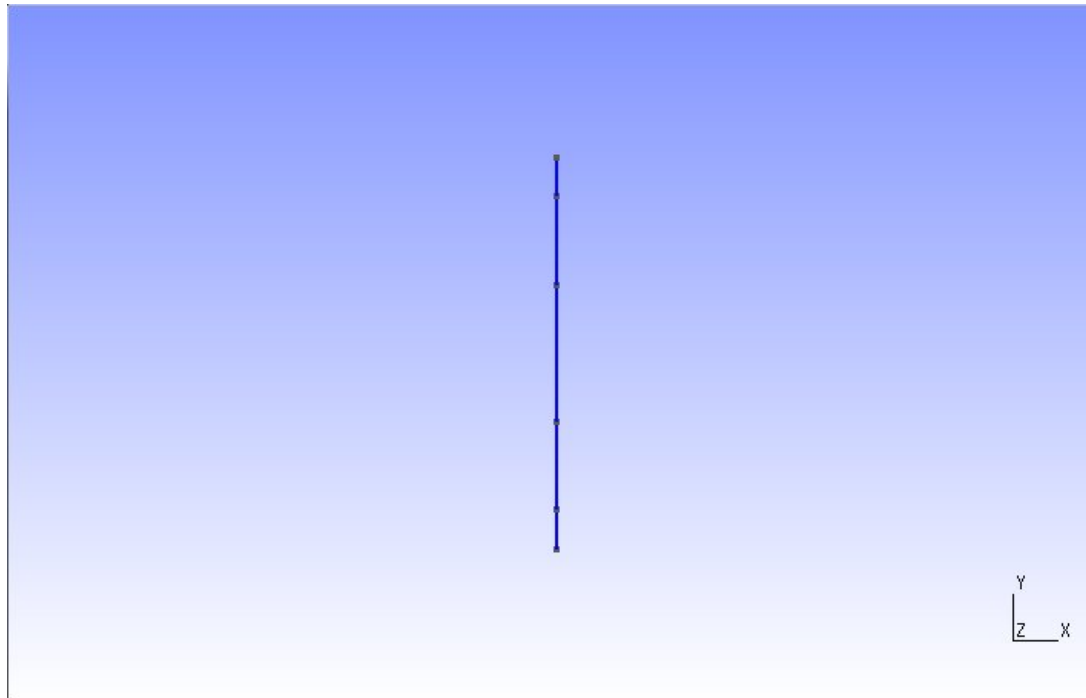
```
Line(2) = {2, 3};
```

```
Line(3) = {3, 4};
```

```
Line(4) = {4, 5};
```

```
Line(5) = {5, 6};
```

❑ Current state



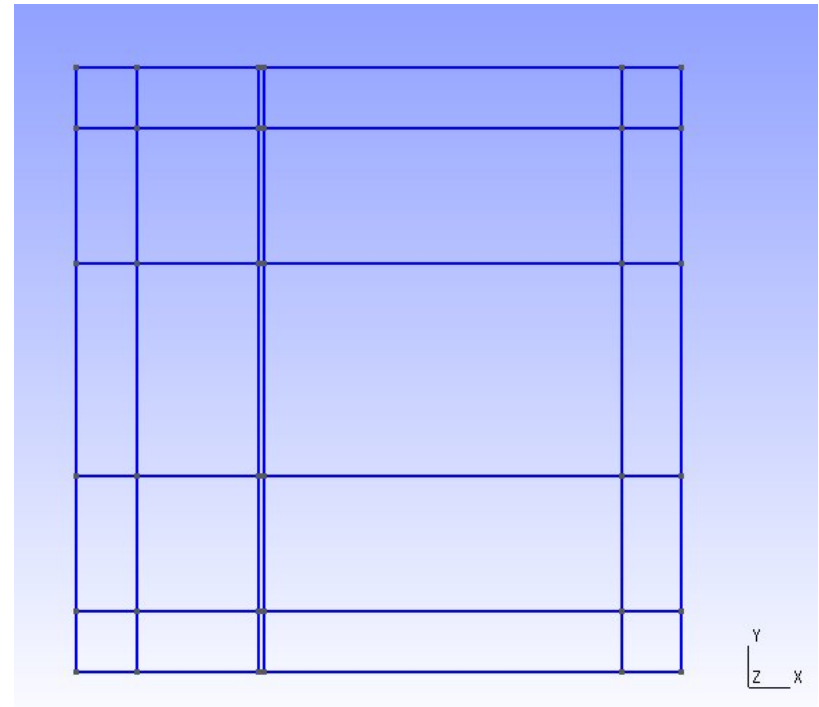
Example 2: Acoustics

- ❑ Extrusion will lead to mis-oriented elements; CFS++ will correct this later on

```
Extrude{ pmlW, 0, 0 }{ Line{1,2,3,4,5}; }  
Extrude{ platePosX, 0, 0 }{ Line{6,10,14,18,22}; }  
Extrude{ plateW, 0, 0 }{ Line{26,30,34,38,42}; }  
Extrude{ overall-2*pmlW-platePosX-plateW, 0, 0 }{ Line{46,50,54,58,62}; }  
Extrude{ pmlW, 0, 0 }{ Line{66,70,74,78,82}; }
```

- ❑ Get rid of multiple entities

Coherence;



Example 2: Acoustics

- ❑ Generate regions to address later in CFS++

```
Physical Surface("prop") = {33,53,73,38,77,41,61,81,37};  
Physical Surface("plate") = {57};  
Physical Surface("pml") = {9,29,49,69,89,93,97,101,105,85,65,45,25,21,17,13};
```

- ❑ Structured meshing and recombine triangles to quadrilaterals

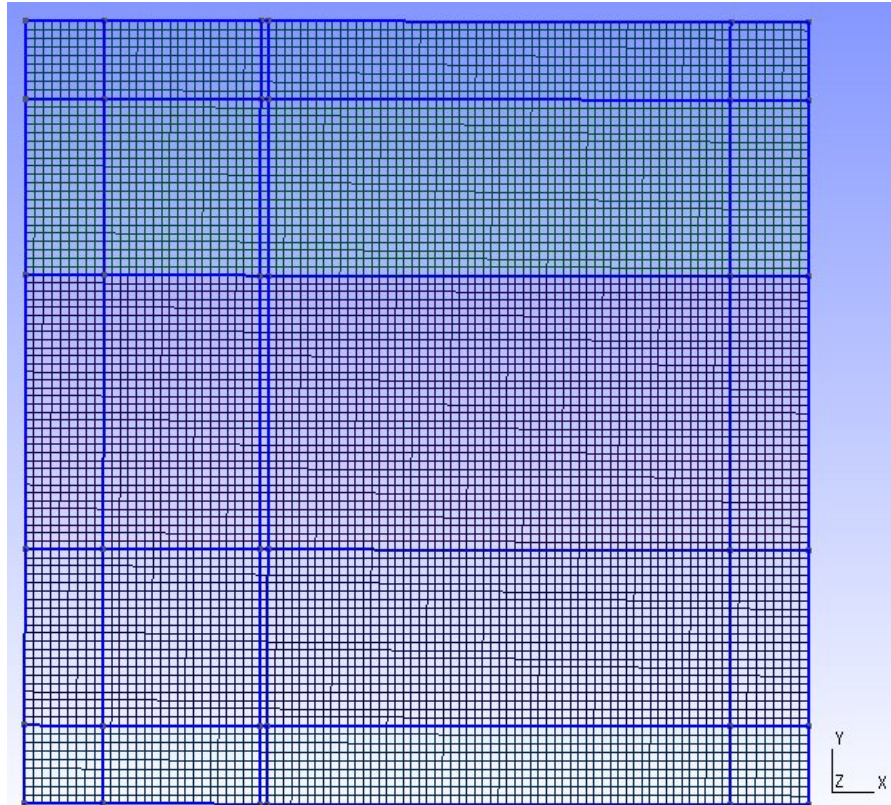
```
Transfinite Surface '*';  
Recombine Surface '*';  
  
//X-Lines  
Transfinite Line {7,8,12,16,20,24,87,88,92,96,100,104} = ((pmlW)/elemSize) Using  
Progression 1;  
.....  
//Y-Lines  
Transfinite Line {1,6,26,46,66,86,5,22,42,62,82,102} = ((pmlW)/elemSize) Using  
Progression 1;  
.....
```

Example 2: Acoustics

- ❑ Perform meshing with 2nd order elements

```
Mesh.ElementOrder = 2;  
Mesh.SecondOrderIncomplete = 1;  
Mesh 2;
```

- ❑ Save the mesh file



Example 2: Acoustics

- ❑ Setup the xml-File for describing the physics and simulation parameter (just will discuss differences to previous example)
 - Hint: search for a similar problem in the CFS++ - TESTSUIT
 - Domain with regions

```
<domain geometryType="plane">
  <regionList>
    <region name="prop" material="air"/>
    <region name="plate" material="air"/>
    <region name="pml" material="air"/>
  </regionList>
  <nodeList>
    <nodes name="excite">
      <coord x="0.0" y="0.0" z="0.0"/>
    </nodes>
  </nodeList>
</domain>
```

Define excitation node

Example 2: Acoustics

❑ Setup the xml-File for describing the physics and simulation parameter

➤ Start the sequence and define analysis

```
<sequenceStep>  
  <analysis>  
    <harmonic>  
      <numFreq> 10 </numFreq>  
      <startFreq> 0.1 </startFreq>  
      <stopFreq> 2 </stopFreq>  
      <sampling>linear</sampling>  
    </harmonic>  
  </analysis>
```

➤ Define PDE for acoustics

```
<pdeList>  
  <acoustic>  
    <regionList>  
      <region name="prop"/>  
      <region name="pml" dampingId="myPml"/>  
    </regionList>
```

Example 2: Acoustics

❑ Setup the xml-File for describing the physics and simulation parameter

- Define the PML (Perfectly Matched Layer) used for free field radiation

```
<dampingList>  
  <pml id="myPml">  
    <type> inverseDist </type>  
    <dampFactor>1.0</dampFactor>  
  </pml>  
</dampingList>
```

- Define excitation and output results

```
<bcsAndLoads>  
  <pressure value="1" name="excite"/>  
</bcsAndLoads>  
<storeResults>  
  <nodeResult type="acouPressure">  
    <allRegions/>  
  </nodeResult>  
</storeResults>
```

Example 2: Acoustics

❑ Material file

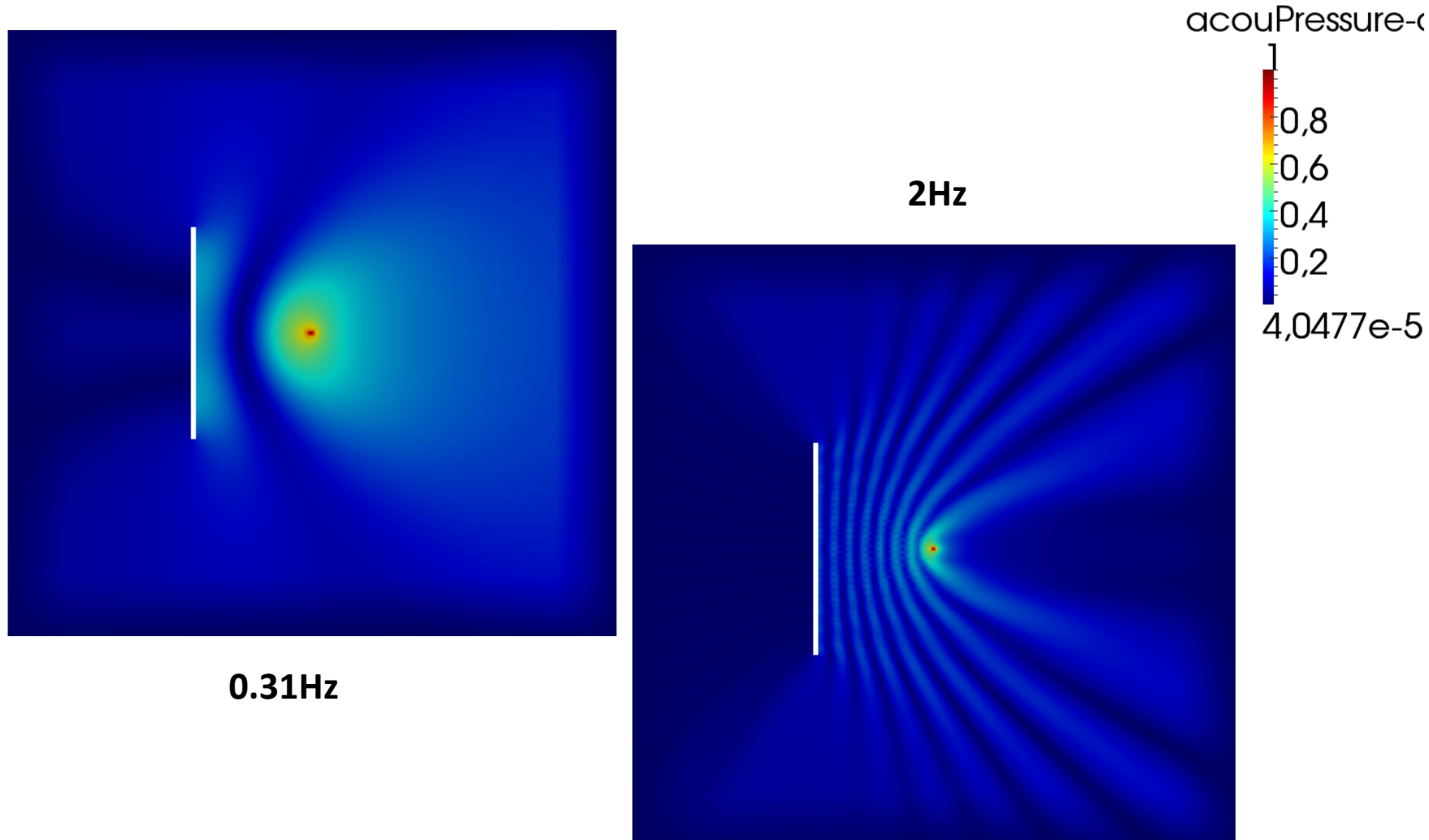
```
<cfsMaterialDataBase xmlns="http://www.cfs++.org">
  <material name="air">
    <acoustic>
      <density> 1 </density>
      <compressionModulus> 1 </compressionModulus>
    </acoustic>
  </material>
</cfsMaterialDataBase>
```

➤ Speed of sound

$$c = \sqrt{\frac{\text{compressionModulus}}{\text{density}}}$$

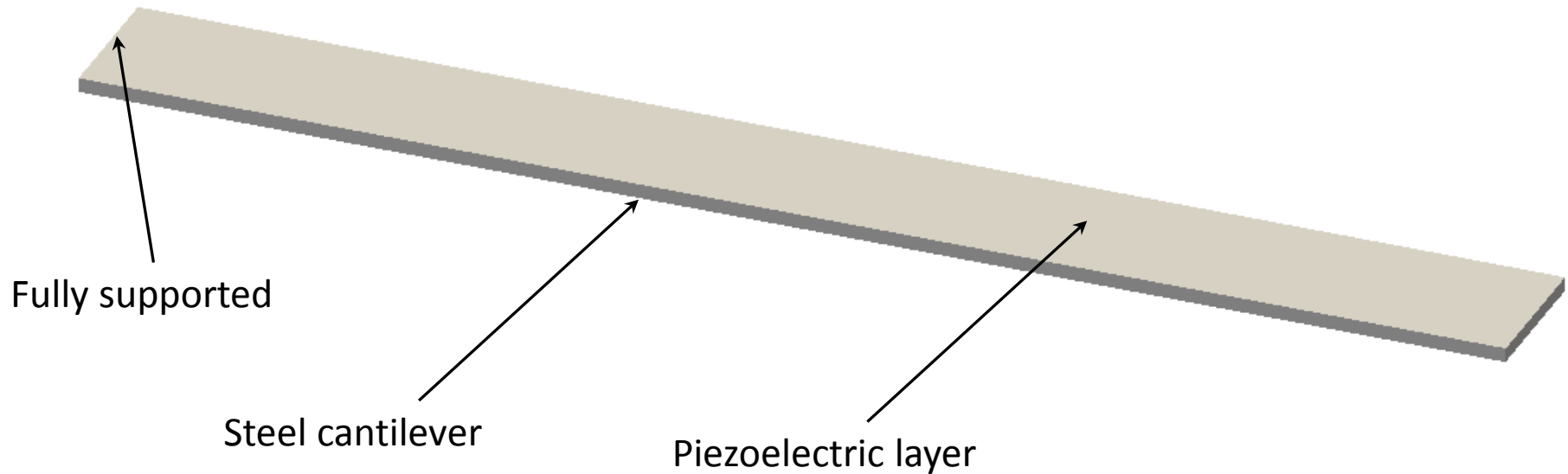
Example 2: Acoustics

- ❑ Perform simulation „cfs noiseBarrier“ and start paraview to investigate in the solution



Example 3: Piezoelectricity

- ❑ Problem setup: Steel cantilever and piezoelectric layer (bimorph)



- ❑ Goal

- Perform a static analysis to obtain deflection, mechanical stresses and electric field intensity

Example 3: Piezoelectricity

- ❑ Preprocessor: Ansys classic, use prep7 commands (just addition parts, which are different to example 1)
 - Generate volume for steel cantilever and piezoelectric layer

```
block,0,L,0,bc,0,b  
block,0,L,0,bc,b,2*b  
nummrg,all,1e-7
```

Cleans the model from doubled keypoints, lines, areas, ..)

- Perform surface meshing for both electrodes (ground and loaded electrode) and for the surface, where the structure is mechanically fixed)

```
setelems,'quadr'  
allsel,  
asel,s,,,2  
asel,a,,,5  
asel,a,,,8  
asel,a,,,11  
amesh,all
```

Example 3: Piezoelectricity

❑ Preprocessor: Ansys classic, use prep7 commands

- Write volume elements for both cantilever and piezoelectric layer

```
allsel,all  
vsel,s,,,1  
eslv  
welems,'cantilever'  
allsel,all  
vsel,s,,,2  
eslv  
welems,'piezoLayer'
```

- Write surface elements for mechanical fixing and electrodes

```
allsel  
asel,s,,,2  
esla  
welems,'gnd'
```

```
allsel  
asel,s,,,8  
esla,,1  
welems,'top'
```

```
allsel  
asel,s,,,5  
asel,a,,,11  
esla,,1  
welems,'fix-all'
```

Example 3: Piezoelectricity

- ❑ Setup the xml-File for describing the physics and simulation parameter (just will discuss differences to previous examples)
 - Domain with regions

```
<domain geometryType="3d">  
  <regionList>  
    <region name="cantilever" material="Steel"/>  
    <region name="piezoLayer" material="PZT-5H"/>  
  </regionList>  
  <surfRegionList>  
    <surfRegion name="gnd"/>  
    <surfRegion name="top"/>  
    <surfRegion name="fix-all"/>  
  </surfRegionList>  
</domain>
```

Example 3: Piezoelectricity

❑ Setup the xml-File for describing the physics and simulation parameter

➤ Mechanical pde

```
<pdeList>
  <mechanic subType="3d">
    <regionList>
      <region name="cantilever"/>
      <region name="piezoLayer"/>
    </regionList>
    .....
  </mechanic>
</pdeList>
```

➤ Electrostatic pde

```
<pdeList>
  <electrostatic>
    <regionList>
      <region name="piezoLayer"/>
    </regionList>
    <bcsAndLoads>
      <ground name="gnd"/>
      <potential value="1" name="top"/>
    </bcsAndLoads>
    .....
  </electrostatic>
</pdeList>
```

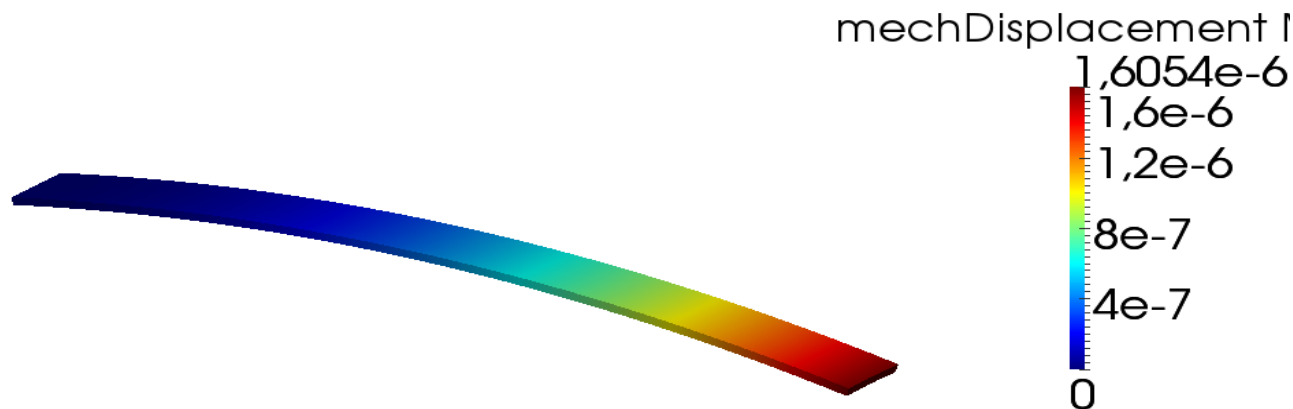
Example 3: Piezoelectricity

- ❑ Setup the xml-File for describing the physics and simulation parameter

- Coupling section

```
<couplingList>
  <direct>
    <piezoDirect>
      <regionList>
        <region name="piezoLayer"/>
      </regionList>
    </piezoDirect>
  </direct>
</couplingList>
```

- ❑ Lets run by typing „cfs bimorph“ and analyse the results by using paraview



Example 3: Piezoelectricity

- ❑ For more details
 - Advanced Pre- and Postprocessing manual
 - XML-Reference manual
 - Introduction to Modeling and Simulation with CFS++
 - Look up the examples in the TESTSUIT

CFS++ – Advanced Pre- and Postprocessing Manual



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